



Figure 1: Architecture of the Qubes OS

To start any application from the console, use the following command:

Syntax: `qvm-run -a <domain> "<app name> [arguments]"`

An example is: `qvm-run -a work python sibi.py`

Syntax to add domains is: `qvm-create`

Syntax to remove domains is: `qvm-remove`

The `qrexec` framework is used by core Qubes components to implement communication between domains. Qubes domains are isolated by design, but there is a need for a mechanism to allow the administrative domain (dom0) to force command execution in another domain (VM).

Some available services of the Qubes OS are:

- Inter-domain file copying
- Dynamic memory management in Qubes
- Disposable VMs
- Dom0 secure update mechanism

Inter-domain file copying

Remote procedure calls (`qrexec` and Qubes RPC) are used for inter-domain file copying. There are two cases when we need a mechanism to copy files between VMs:

- 'Regular' file copy, when a user instructs a file manager to copy some given files/directories to a different VM.
- DispVM copy, when a user selects 'Open in DispVM' on a file; this file must be copied to a disposable VM, edited by the user and, possibly, a modified file must be copied back from DispVM to the VM.

Dynamic memory management in Qubes

Qmemman (the Qubes memory manager) is used for memory management. Qmemman checks for the 'do-not-membalance' file before allocating memory for the new domain. If it finds the file, it suspends memory balancing. The command to free the specified amount of memory is:

```
/var/run/qubes/qmemman.sock
```

Xenstore is used for VM memory management. Traditionally, Xen VMs are assigned a fixed amount of memory. It is not the optimal solution, as some VMs may require more memory than that assigned initially, while others

Name	State	CPU	MEM
dom0	Running	17%	1848 MB
netvm	Running	0%	400 MB
firewalld	Running	0%	264 MB
itlvpn	Running	0%	264 MB
work-pub	Running	0%	783 MB
blog	Running	0%	565 MB
work	Running	0%	695 MB
Personal-admin	Running	0%	325 MB
qubes	Running	0%	1000 MB
personal	Running	1%	1173 MB
work-admin	Running	0%	304 MB
vault	Running	0%	271 MB

Figure 2: Qubes virtual manager

under-utilise memory. Thus, there is a need for a solution capable of shifting free memory from one VM to another.

Tmem provides a pseudo-RAM that is assigned on an 'as per-need' basis. Basically, Tmem is integrated with the Linux kernel and provides an interface to copy memory to/from fast (RAM based storage).

Disposable VMs

In general, disposable VMs are lightweight; they are created and booted in a short period of time (less than one second) with the sole purpose of hosting only one application.

The Dom0 secure update mechanism

In general, Dom0 executes a secured `qvm-dom0-update` script under `updateVM`, where RPM files are checked, verified and downloaded. Once all the updates are downloaded, the update script requests for RPM service `qubes.ReceiveUpdates` to be executed in Dom0. This service is implemented by the `qubes-receive-updates` script running in Dom0. The Dom0's `qvm-dom0-update` script waits until `qubes-receive-updates` has finished. The `qubes-receive-updates` script processes the untrusted input from `Update VM`; it first extracts the received `*.rpm` files (that are sent over the `qrexec` data connection) and then verifies the digital signature on each file in order to avoid processing compromised updates. The `qubes-receive-updates` script is a security-critical component of the Dom0 update process. **END**

References

- [1] <https://www.qubes-os.org>
- [2] <https://www.qubes-os.org/doc/qubes-architecture/>
- [3] <https://www.qubes-os.org/getting-started/>

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